TKH M1 Report: temporal semantic abstraction

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1 Problem, data and explicit trade-off

The task is to organize an evolving indexed knowledge hypergraph into nested, interpretable groups while preserving its original assertions. The supplied Collection 10 has 5798 entities, 1429 indexed hyperedges and 8929 incidences. IDs distinguish even equal-support assertions. Nodes are admitted first under the recorded source-availability policy; edges require admissible dates/status and admitted endpoints. There are 304 unresolved nodes and 158 quarantined edges in normalization. These are preserved, not silently repaired.

Snapshot	Nodes	Edges
2020	1424	352
2022	2093	504
2024	4046	912

The hierarchy fits $K_0 = \min(12, n)$ coarse blocks first, fixes their boundaries, then fits $\min(4, |C|)$ children inside each parent C. Thus $K_1 \le 48$; the terminal layer consists of n singletons. All three selected snapshots realize 12 and 48 groups. This sequential construction satisfies laminarity without claiming a joint optimum over levels.

For one partition P of the admitted nodes, the criterion is

$$
\begin{aligned}
J(P; P^-) &= (1-\alpha)S(P) + \alpha F(P) + \beta T(P, P^-), \\
S(P) &= A^{-1} \sum_{C \in P} \sum_{v \in C} a_v \|x_v - \mu_C\|^2, \quad \mu_C = \frac{\sum_{v \in C} a_v x_v}{\sum_{v \in C} a_v}, \\
F(P) &= W^{-1} \sum_{e : r_e \ge 2} w_e \frac{\lambda_e(P) - 1}{r_e - 1}, \\
T(P, P^-) &= 1 - |U|^{-1} \max_{M} \sum_{(i,j) \in M} O_{ij}.
\end{aligned}
$$

Here x_v is the frozen MiniLM surface-form vector, $a_v > 0$ is node mass (one in this corpus), $A = \sum_v a_v$, r_e is original arity, and λ_e counts occupied clusters. Relation balancing sets $w_e = 1/c_\rho$ for c_ρ active nontrivial edges of type ρ, with $W = \sum w_e$. U is the persistent-node set; O_{ij} counts its overlaps and M is a one-to-one matching. Set $F = 0$ if $W = 0$ and $T = 0$ if U is empty. The selected $\alpha = 0.75$, $\beta = 0.05$ give $J = 0.25S + 0.75F + 0.05T \le 1.05$ for unit/bounded vectors.

Construction-only development used 30 alpha fits and 60 synthetic beta-calibration fits. No final MPNet, NLI or retrieval score selected coefficients. An alternative is pairwise projection followed by graph clustering: it is convenient but changes the native connectivity functional. Leiden, native partitioning, spectral reduction and multislice methods motivate comparisons; their guarantees do not transfer to this objective [1–4].

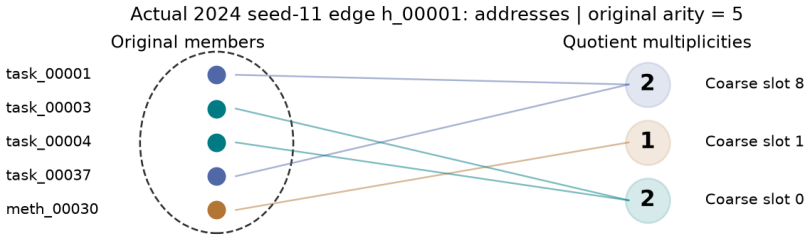
2 Native accounting, algorithm and guarantees

Let $B \in \{0,1\}^{n\times m}$ be the node–edge incidence matrix and $\mathsf{S} \in \{0,1\}^{n\times K}$ the partition membership matrix. The quotient is

$$M = \mathsf{S}^{\mathsf{T}} B, \qquad M_{Ce} = |C \cap e|, \qquad \sum_{C} M_{Ce} = r_e.$$

For fine-to-coarse parent membership R, $\mathsf{S}_0 = \mathsf{S}_1 R$ gives $M_0 = R^{\mathsf{T}} M_1$. Production storage is transposed, edge by cluster. Every original edge ID, provenance reference, internal edge and multiplicity is retained. Equal quotient support does not merge distinct assertions.

For an arity-four edge, complete collapse (4), partial collapse $(2,2)$ and a three-way split $(2,1,1)$ produce native costs $0, 1/3, 2/3$ before weighting. The partial split has pairwise cut fraction $2/3$: projection and native connectivity measure different preferences. Exact multiplicity recovery does not guarantee preserved spectra or semantic truth.



Semantic or incidence initialization and feasible warm-start repair precede refinement. Coarse fitting uses all active edges; child moves stay inside their fixed parent, but child structural accounting still includes cross-parent edges. A legal single-node move cannot empty its source. Maintained masses, vector sums and integer incidence counts yield exact algebraic gains for the declared fixed matching. A SciPy rectangular assignment refreshes the matching at each sweep boundary [5]. With M_s optimal at sweep start,

$$J(P_{s+1}) \le \widetilde{J}(P_{s+1}, M_s) \le \widetilde{J}(P_s, M_s) = J(P_s).$$

This is a sweep-boundary argument; it does not promise rematched descent after every intermediate move. The move tolerance is 10^{-10} and the cap is 30 sweeps. Full recomputation checks and the randomized checkpoint remain in the optimizer stream. A no-move sweep establishes fixed-current-matching single-node stationarity only. One latest diagnostic level reaches the cap; the main conditions retain their actual stop records.

A conservative per-level numerical-work account is

$$O\Big(s\,[K(nd+L) + K_*^3 + mK + Kn]\Big).$$

Here s is sweeps, d vector dimension, L original incidences, m edges, and K_* bounds the assignment dimension. Dense mK counts, nd vectors, L incidence lists and cluster moments require storage. Initialization, repair, I/O, model inference, repeated evaluation and report/package builds add separate costs.

Established by construction/checks	Empirical or not established
Partition coverage, nonempty quotas, parent containment	Scientifically unique or uniformly meaningful groups
Original-ID/multiplicity preservation	General query or spectral equivalence
Fixed-surrogate gain and boundary nonincrease	Global optimum or rematched single-move optimality
Pinned cached replay contract	Universal cross-platform byte identity

3 Controlled evaluation and observed trade-offs

The matrix contains 75 base, 45 paired 10% edge-removal, 15 warm-zero and 15 uniform-weight conditions over five seeds. The selected trajectories are a subset, not extra replicates. Five latest-view diagnostic fits bring the total to 155. Perturbation retains the entire node universe, paired streams and the specified predecessor. All 15 semantic-only perturbation controls are invariant.

MPNet, withheld from construction and selection, measures nonzero-vector pair cosine [6]. This is an independent instrument choice, not independent pretraining or domain truth. Macro and pair-weighted averages are distinct. The 180 cells include 99 size/evaluability-preserving assignment nulls each (17,820 scores); null spread is not a population confidence interval.

2024 coarse	Cosine	Null mean	Native F
Semantic	0.272	0.116	0.457
Structure	0.225	0.116	0.000
Joint static	0.226	0.116	0.155
Joint temporal	0.231	0.116	0.147
Pairwise temporal	0.187	0.116	0.094
Warm beta=0	0.242	0.116	0.138

The structure-only preference can yield a dominant large group; heterogeneous type mixtures remain in the error analysis. Connectivity depends on the number of occupied groups, not how evenly endpoints are distributed among them. Hard counts do not impose balance or scientific purity. Adding a balance penalty now would change the registered method.

Adjusted Rand index (ARI) compares partitions on the same common-node universe [7]. Real transitions and edge-removal stability answer different questions. The following coarse transition results contrast the temporal penalty with a warm-start control that already contains historical information.

Transition	Joint temporal	Warm beta=0	Difference
2020-2022	0.995 [0.992, 0.998]	0.901 [0.883, 0.919]	0.094 [0.077, 0.111]
2022-2024	0.988 [0.981, 0.994]	0.840 [0.802, 0.878]	0.148 [0.108, 0.188]

Entries give mean [approximate 95% interval] over five trajectories; paired differences use the same seeds. The interval is $\bar{x} \pm t_{.975,4} s_x/\sqrt{5}$, conditional on this corpus and run mechanism. High temporal agreement does not show that suppressed changes were scientifically undesirable. Terminal singleton agreement is automatic, not learned stability. Full fine-level, seedwise and locality records are bundled.

The source-attributed timeline is retrospective. Literal name occurrences, exported relationships and adjudicated entailment are different evidence classes. Modern frozen encoders may contain later knowledge: these experiments are not predictions made by a pre-2020 model. Candidate identity conflations remain unresolved overlays; no raw entity remapping was performed.

4 Descriptions, evidence and retrieval failures

The final deterministic lexical/extractive policy nominates TF–IDF themes but independently checks contiguous phrases in original normalized member names. For $m > 0$ substantive members, multiword themes need $\min(m, 2)$ exact occurrences; unigrams need $\min(m, \max(2, \lceil .04m \rceil))$. Verified, cutoff-eligible source support is also required. A one-member content pool is explicitly described as such; bibliography-only groups receive no invented substantive theme. All 876 outputs are retained.

Coarse (465 members). Mixed topics: Neural Network and Equivariant graph. This mixed group includes Neural Network and Equivariant graph; a cited source states: "Here we present the virtual node graph neural network to address the challenges". Exact lexical coverage: 9.5%.

Child (453 members). Mixed topics: Neural Network and Equivariant graph. This mixed group includes Neural Network and Equivariant graph; a cited source states: "Here we present the virtual node graph neural network to address the challenges". Exact lexical coverage: 9.7%.

Bibliography-dominated (607 members). Mixed bibliography: method and dataset records. This bibliography-dominated group (72% author/citation records) collects records on Ice Ih and JAX MD, chiefly author and cited work; eligible excerpts support these topics but do not establish one shared mechanism. Exact lexical coverage: 0.8%.

The examples use the first eligible construction slot and its first child, plus the most bibliography-dominated coarse group, rather than choosing by evaluator score. Repeated coarse/child wording shows that a quota alone need not create a new semantic resolution. Quotes belong to the cited source, not every cluster member.

Selected descriptions	Baseline	Final
Mean member cosine	0.217	0.289
Own-centroid top-1	0.494	0.583

Across 180 selected responses, mean exact lexical coverage is 22.6%. These finite-corpus measurements are not human informativeness ratings. In the 60-response selected NLI sample, 44 responses have no substantive source claim. The assessed-claim nonsupport ratio is 0.767 [0.558, 0.977], based on 16 sampled claims. Inverse-inclusion weighting and finite-population ratio linearization condition on the automated verdicts [8]; they do not cover model error. The uncalibrated classifier matches 6 of six synthetic domain controls and rejects many exact quotations.

All 720 exposed-question cells remain recorded: 130 measured and 590 unavailable. Nine node and four evidence questions are measurable in the latest view. Both systems have zero top-1/5/10/20 recall. A registered larger-budget evidence result coexists with that failure:

Evidence budget	Flat recall	Hierarchy recall
50	0.000	0.000
100	0.000	0.062
200	0.125	0.312

A budget charges route summaries and examined answers, not all dot products; actual vector-scoring work is recorded separately. Four evidence questions, partial gold and exposed answers prevent a general superiority claim. Neither route supports reliable scientific question answering.

5 Reproduction, limits and continuation

The backend, original data, exact task excerpts, fixed vectors, input-keyed annotations, active results, tests, LaTeX sources, static hierarchy figures and static reader are supplied together. The author campaign took 663.9 seconds; summed fit wrappers took 265.2 seconds. These are measured run scopes, not model-acquisition or source-adjudication costs. Detailed phase records retain unmeasured costs as unavailable.

The supported contract is exact cached scientific replay on the recorded Python 3.12/macOS arm64 pinned environment. The recipient run starts with absent fit storage, reconstructs 150+5 fits, regenerates labels/evaluation, and compares 930 fit references and deterministic records. Model inference and full-paper extraction are separate optional regeneration tasks. Installation history, current commands, exit codes and comparisons are indexed in `release/payload_acceptance.json` and its logs; a manifest binds their immutable payload. The earlier Linux annotation comparison failure remains historical evidence of a portability limit, despite exact numerical-fit reproduction. A toleranced annotation checker is optional and does not replace the strict acceptance contract.

Limits. One sparse corpus, surface-form embeddings, conservative dating, unresolved identity and source entailment, sequential local optimization, mixed/bibliographic groups, limited lexical coverage, five seeds, uncalibrated NLI and exposed incomplete questions constrain interpretation. No external submission, human peer-review certification, global optimum, calibrated truth judgment or universal byte reproducibility is asserted.

Four-week continuation. Week 1: adjudicate dated assertion spans and identities, and freeze unseen questions. Week 2: evaluate prefix-safe contextual representations with controlled ablations. Week 3: compare richer native moves and additional seeds, retaining costs and negative controls. Week 4: obtain blinded domain-expert abstraction/evidence judgments and replicate on a second corpus. These are prospective studies, not completed experiments.

AI disclosure. Codex substantially assisted implementation, prose, tests and orchestration; AI-assisted review is not independent human review. `AI_USAGE.md` documents AI assistance and identifies ten reconstructed prompt examples. The static reader displays the recorded results without fitting. The separate mathematical formalism provides full derivations and the technical supplement; the algorithm documents execution; the present report contains the central method, guarantees and negative results.

References

[1] V. A. Traag, L. Waltman, and N. J. van Eck. From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports*, 9:5233, 2019. doi: 10.1038/s41598-019-41695-z.

[2] Sebastian Schlag, Tobias Heuer, Lars Gottesbüren, Yaroslav Akhremtsev, Christian Schulz, and Peter Sanders. High-quality hypergraph partitioning, 2021.

[3] Andreas Loukas. Graph reduction with spectral and cut guarantees. *Journal of Machine Learning Research*, 20 (116):1–42, 2019. URL https://jmlr.org/papers/v20/18-680.html.

[4] Peter J. Mucha, Thomas Richardson, Kevin Macon, Mason A. Porter, and Jukka-Pekka Onnela. Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328:876–878, 2010. doi: 10.1126/science.1184819.

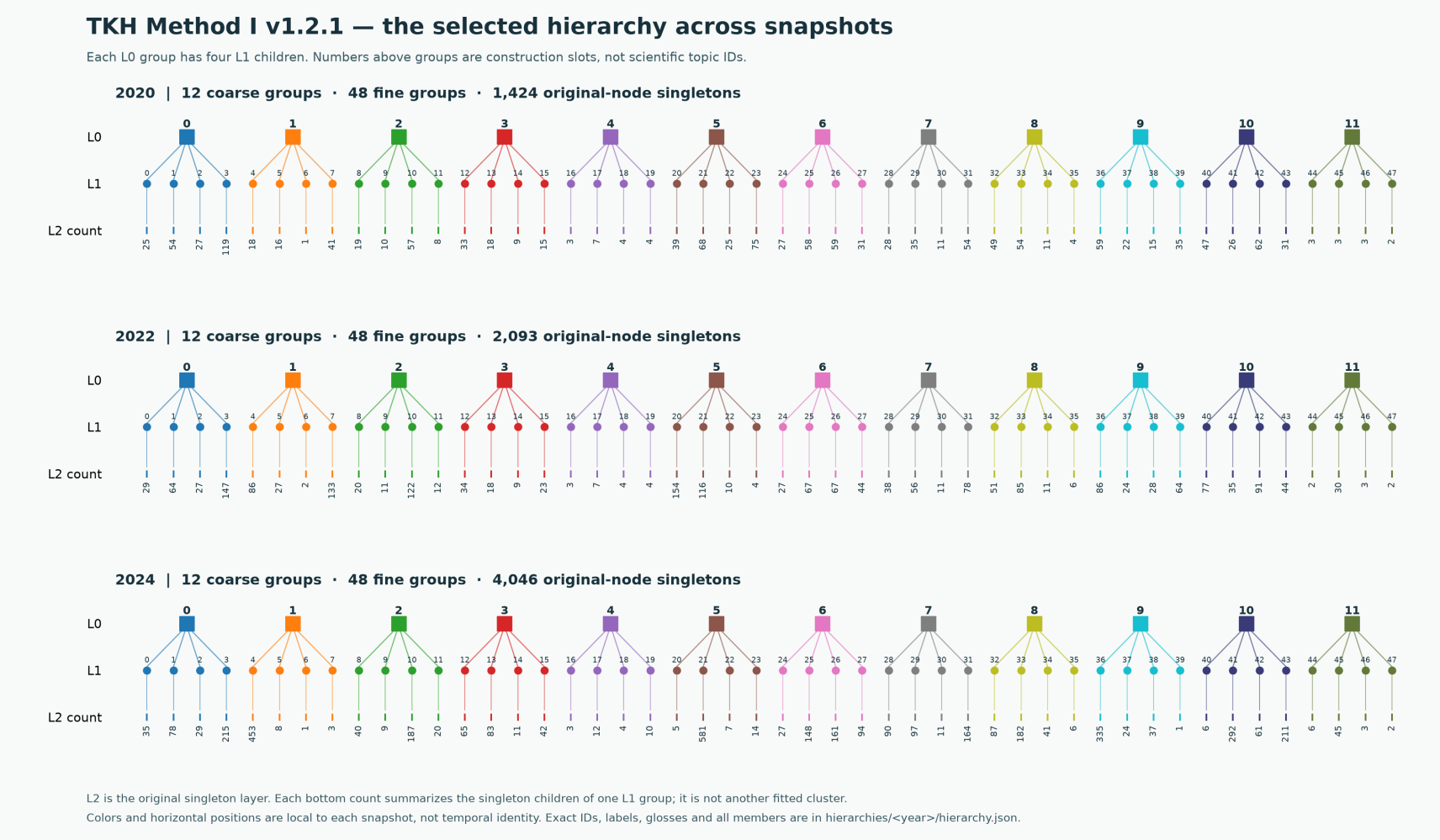
[5] SciPy developers. linear_sum_assignment: rectangular linear assignment. SciPy reference documentation, 2026. URL https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html. Consulted 11 October 2026; installed version recorded in acceptance.

[6] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks, 2019.

[7] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075.

[8] D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47:663–685, 1952. doi: 10.1080/01621459.1952.10483446.

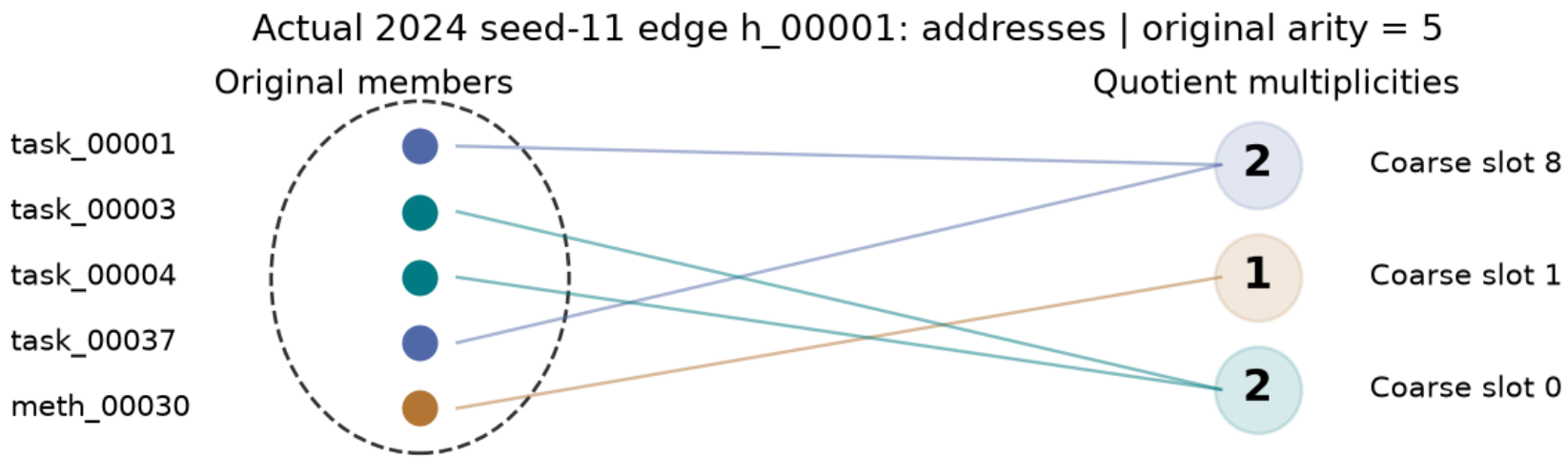
Hierarchy across the three selected snapshots



Each snapshot contains 12 coarse groups (L0), 48 fine groups (L1), and the original singleton layer (L2). The bottom numbers are singleton counts under each L1 group, not extra fitted clusters. Colors and horizontal positions are local to a snapshot, not temporal identity. The original high-resolution image is embedded; zoom to inspect dense slot labels.

Source: `visualization/levels_0_1_2.png`. Exact IDs, parent links and members: `hierarchies/<year>/hierarchy.json`.

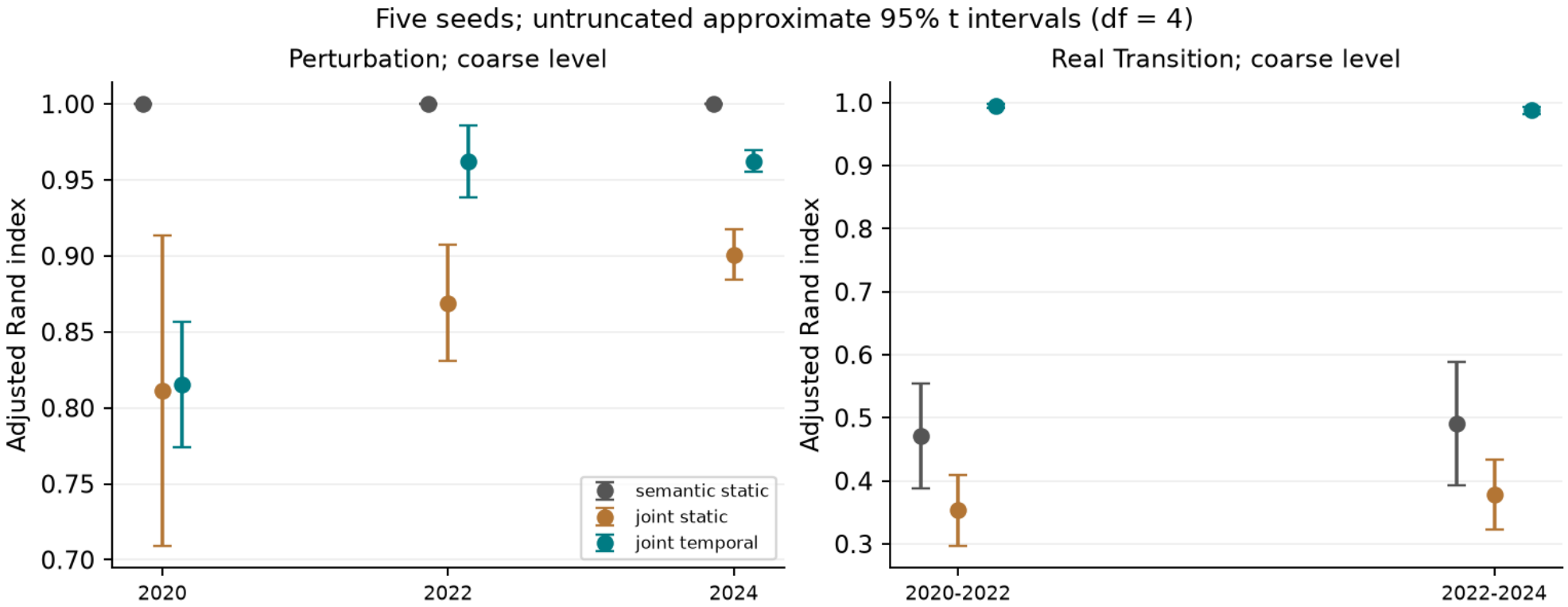
Native hyperedge coarsening with conserved multiplicities



Expanded view of the figure on assessment page 2. The actual selected 2024 seed-11 hyperedge has five original members. They occupy three coarse groups with multiplicities 2, 1 and 2, whose sum remains five. This is an incidence-preserving quotient of one indexed assertion, not a pairwise replacement graph or a claim of scientific entailment.

Source: `report/figures/native_collapse.png`; exact provenance and membership: `report/figures/native_collapse_source.json`.

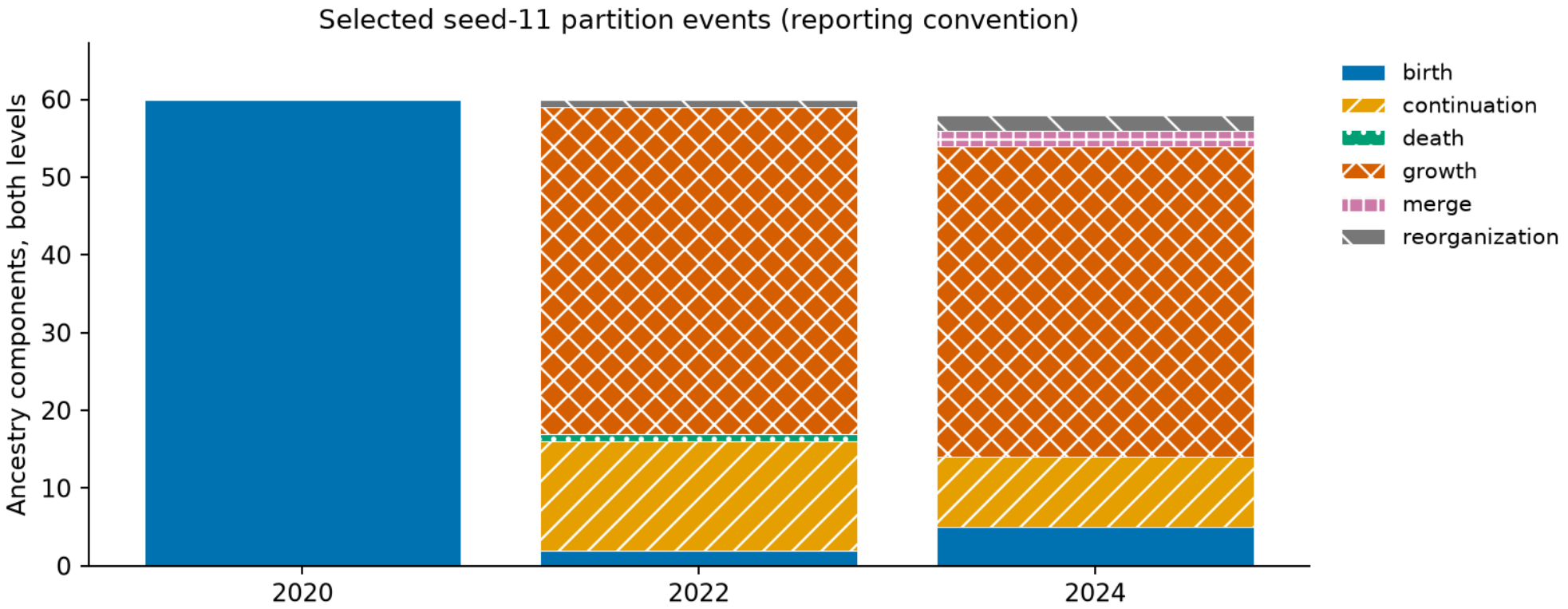
Stability under perturbation and across real transitions



The left panel compares coarse partitions after the registered current-edge perturbation; the right panel compares actual chronological transitions. Points are five-seed means with untruncated approximate 95% Student t intervals (four degrees of freedom). These intervals describe the declared run mechanism on this corpus; high agreement does not establish scientific correctness.

Source: `report/figures/stability_intervals.png`; underlying estimates and intervals: `results/stability_intervals.json`.

Recorded temporal events in the selected trajectory



The stacked bars count ancestry components at both fitted levels in the selected seed-11 trajectory. The complete log contains 178 events: 67 births, 82 growth events, 23 continuations, three reorganizations, one death and two merges. No split events were observed. These categories are reporting conventions for partition ancestry, not scientific discovery or extinction.

Source: `report/figures/event_flow.png`; exact event records: `temporal_events.json`. The PNG and SVG files in the two figure folders are alternate formats of the same four images, shown once each in this appendix.